Measurement inputs for two additional manuscripts

Human Compression Project

11 October 2026

This packet prepares two independent comparisons. It measures the complete source manuscripts and the complete replacement papers, including their common theories. It supplies no compression factor and no new mathematical credit. The accepted scientific contracts and their separate reviews remain the authority for the recovered results. The two originals are F015 and L028; their 64 detailed source records are not 64 manuscripts.

Complete documents

The primary coordinate is NFC characters excluding whitespace, after the explicitly recorded glyph calibration and removal of repeated page furniture. Titles, abstracts, tables, printed software, references, acknowledgements and author information remain. Each of the following costs belongs to one complete document.

Document	Pages	Primary count
F015: original matching-extension article	21	37,951
Matching: full common theory	6	16,732
Matching: full application	7	17,168
Matching: full program appendix	2	4,998
L028: original chiral-enumeration article	14	29,599
Chiral enumeration: full common theory	4	8,831
Chiral enumeration: full application	9	23,171
Chiral enumeration: full program appendix	5	11,940

F015 is the complete final journal edition of Piedade's *Infinite Families of Hypertopes from Centrally Symmetric Polytopes*, EJC 30(2) (2023), P2.20. Its first-page DOI and journal identification remain; only the repeated later footers are removed. L028 is the complete fourteen-page revised ULB author manuscript of Buekenhout, Leemans and Tranchida, *A new algorithm to classify chiral polytopes with a given automorphism group*. It is neither the earlier arXiv version nor the unavailable journal typesetting. Its complete printed Magma code, numerical table, references and author addresses remain.

The eight PDFs contain 68 pages. Thirty coordinates are supplied for every document: five units under NFC and NFKC, in raw untrimmed, raw selected and calibrated selected text. The units are whitespace tokens, alphanumeric runs, nonwhitespace characters, characters after collapsing whitespace and UTF-8 bytes after collapsing whitespace. They measure extracted text, not mathematical difficulty or the full visual content of a page.

Programs and formal evidence

The matching appendix prints the entire exceptional-orbit GAP program. It is a finite premise of the exceptional-family proof. The chiral appendix prints the entire reference enumeration, exact formula-evaluation script and independent GAP involution-count validation. Those finite tests do not replace the written theorem for arbitrary finite groups and all ranks.

The raw whole-file counts below are an alternative to printed program costs; they are never added to the printed appendix a second time.

Complete file	NFC nonwhitespace characters
Matching exceptional-orbit program	4,408
Chiral reference enumeration	8,580
Chiral table evaluation	1,421
Chiral GAP validation	1,089

All three operational drivers are recorded separately, in full, as are the two Lean files. The fourteen Lean declarations verify specified group identities and coordinate transport. They do not formalize the complete geometry, all finite algorithms, or every retained outside theorem. No scientific program or Lean proof is rerun by this measurement packet; their earlier independent acceptances are bound explicitly.

Encoding, scope and limitations

All source Type1 text events are aligned to actual PDF font dictionaries, encoded bytes and embedded glyph outlines. The two chiral mathematical papers and the new program appendix receive the same audit for their legacy Type1 mathematical glyphs; their Unicode Type0 body text is unchanged and is not claimed to have a separate outline audit. The audit records 78 Type1 font dictionaries and 1,358 glyph classes. There are 660 occurrence-bound replacements in 52 classes, with actual page, byte, font, event and context evidence. All 68 pages and 88 exceptional or repaired glyph classes were visually inspected.

Only visible glyph identity is calibrated. A maps-to symbol assembled from a bar-and-stub glyph and an arrow remains two components; it is not collapsed to one character. A display-wrap hook in the new code appendix likewise has two glyph pieces. Its left hook has no asserted exact standalone Unicode equivalent and retains the literal extracted comma. That representation limit particularly affects UTF-8 comparisons. All wrap markers remain in the measured PDF costs; they are ignored only for the independent check that every literal program character is printed. No mathematical correction is made to either source text.

The matching contract retains the original extension and lattice theorem and the stated coordinate descriptions of toroids. The complete intended finite families are recovered with their explicit source corrections: the printed 120-cell presentations omit twenty unordered-pair orbits and define infinite groups. The corrected finite presentations and the literal failure are kept distinct. Incidental historical catalogue comparisons are outside this contract.

The chiral contract retains the Schulte–Weiss geometric reconstruction and the Liebeck–Shalev estimate with attribution. Oriented and unrooted classes are distinguished; duality is not quotiented. The source's timings remain empirical observations. Correcting numerical table entries and the regular-output boundary does not create a new original manuscript or an open-question solution.

For later comparisons, each topic must pay for its full common paper and its complete application. Conditional application costs assume that the entire common paper and the complete declared program bundle have already been acquired. Shared material is not free across the two topics. Full retained external proofs, original software not included in the source PDF, full graphic and layout costs, and end-to-end formalization costs remain unknown, not zero. F015 contains vector diagrams; the absence of bitmap images does not remove those costs.